

Stock Market Index Direction Prediction Using Limit Order Book Data

EAI format — CONTENT PREVIEW (final PDF: Overleaf pdfLaTeX)

Joseph Raj, Sumanth G. M, Abhishek Kumar, Kumara Swamy, Raghav, Saagar

Dept. of Data Science, ML & AI, PES University, Bengaluru, India

Abstract

Abstract. Short-horizon prediction of mid-price direction from Limit Order Book (LOB) data is a central problem in market microstructure, but research is concentrated on developed markets and the public FI-2010 benchmark. We address two gaps. First, a *data-engineering* gap: because no clean Indian index-futures LOB dataset existed, we reverse-engineered the undocumented Dhan twenty-level binary depth feed, built a cloud collection pipeline (an EC2 WebSocket collector writing daily Parquet to Amazon S3) with automated token refresh and contract-expiry handling, and produced a continuous front-month series for NIFTY and BANKNIFTY index futures (≈ 1.1 million events per instrument) aligned to the FI-2010 contract. Second, a *modelling* gap: we introduce **MambaLOB**, a selective state-space (SSM) architecture with linear-time $O(L)$ scaling, and a convolutional hybrid **ConvMambaLOB**, and compare them against reproduced DeepLOB, MLPLOB and TLOB baselines under one protocol. We further engineer microstructure features (order-flow imbalance, micro-price, depth imbalance, and a broker-unique order-count channel) and validate every claim with multi-seed confidence intervals, paired significance tests, calibration, and hyperparameter optimisation. The headline result is reported honestly: the SSM models recover roughly 90% of the transformer’s accuracy at $12\text{--}25\times$ fewer parameters — an efficiency result rather than an accuracy win — while microstructure feature engineering, not architecture, drives the only statistically significant improvement on the Indian market.

Keywords: Limit Order Book, Mid-price Prediction, State-Space Models, Mamba, Market Microstructure, NSE Index Futures, Deep Learning, Data Engineering

1 Introduction

Electronic exchanges operate as continuous double auctions whose state is recorded in a Limit Order Book (LOB): the set of outstanding bid and ask orders at each price level. The LOB carries a rich microstructural signal for short-horizon price prediction, and deep learning models have shown small but real predictive power on it. However, this research is concentrated on developed markets (NASDAQ, Nordic exchanges) and on the single public benchmark, FI-2010 [5]. Indian index futures are highly liquid yet structurally distinct (tick discreteness, regime shifts, distinct liquidity profiles) and remain almost entirely unstudied with modern LOB architectures.

Index direction is hard to predict because the series is non-stationary and governed by rapidly evolving supply-demand dynamics. The order book updates irregularly and at high frequency, yet most prior deep LOB models train on coarse fixed-grid snapshots (e.g. one-second sampling) that lose intra-interval dynamics. We capture NSE depth at the broker’s native push cadence (≈ 2.5 Hz), finer-grained than typical one-second snapshots, while noting transparently that this remains a near-regular snapshot feed rather than tick-by-tick data.

This paper makes the following contributions:

- **An engineered NSE index-futures LOB dataset.** To our knowledge the first ML-ready Indian index-futures LOB dataset, built end-to-end from the reverse-engineered Dhan binary feed

through a documented cloud pipeline and a front-month stitching/preprocessing pipeline aligned to FI-2010.

- **MambaLOB and ConvMambaLOB.** A selective state-space architecture for LOB direction classification and a convolutional hybrid, with a rigorous efficiency comparison ($O(L)$ vs. $O(L^2)$).
- **Microstructure feature engineering.** Engineered order-flow imbalance, micro-price, depth imbalance, and a broker-unique order-count channel, shown to give a statistically significant gain.
- **An honest transfer and rigour study.** A standardized comparison of four architectures on the NSE benchmark with multi-seed significance, calibration, and hyperparameter optimisation, and transparent reporting of what does and does not transfer from FI-2010.

2 Related Works

2.1 Classical techniques

Parametric time-series models (AR, ARMA, ARIMA) and Hidden Markov Models were first applied to price forecasting. They model only the derived price/return series, ignore book depth, and fail under the non-linear, non-stationary behaviour of high-frequency LOB data; they serve only as conceptual baselines.

2.2 The FI-2010 benchmark and CNN–LSTM models

FI-2010 [5] is the first public LOB benchmark — ten levels of depth for five Nasdaq Nordic stocks over ten days — providing normalised feature variants, a smoothed mid-price labelling rule at horizons $k \in \{10, 20, 50, 100\}$, and standard cross-folds. DeepLOB [8] is the seminal deep model: a convolutional stack that fuses price/volume then book sides, an Inception module for multi-scale temporal patterns, and an LSTM. It established the interleaved 40-feature $\times$ window representation that all subsequent work (including ours) adopts, but is limited by its sequential LSTM and lack of attention.

2.3 Transformer and state-space models

MLPLOB and TLOB [1] reframe LOB models as stacks of normalization-plus-mixing blocks; TLOB uses *dual* (temporal and feature) self-attention and is the current state of the art on FI-2010. Structured state-space models — S4 [4] and the selective-scan Mamba [3] — offer sub-quadratic and linear-time sequence modelling competitive with transformers. Their use as a discriminative classification backbone for LOB direction is the gap this work targets.

2.4 Benchmark studies and microstructure priors

The LOBCAST benchmark study [6] re-implemented fifteen deep LOB models and found they overfit FI-2010 and degrade sharply on unseen data — motivating our honest transfer framing and dual-metric reporting. Independently, the microstructure literature supplies strong priors: order-flow imbalance is a robust predictor of short-horizon moves [2], and the micro-price [7] is a volume-weighted fair-value estimator superior to the mid. These underpin our engineered features.

3 Methodology

3.1 Engineered NSE dataset and data pipeline

Because no clean Indian index-futures LOB dataset existed, the dataset was engineered end to end. The Dhan twenty-level depth WebSocket delivers an *undocumented binary frame*, which we decoded offline: a 1992-byte full format (a 40-byte header followed by twenty 16-byte bid and twenty 16-byte ask records of order-count, price, and quantity) and a 664-byte compact format. Decoding required fixing non-obvious issues — the bid price sits at byte offset +4, empty levels emit denormalised near-zero floats that must be rejected, and quantity/order-count fields overflow 32-bit integers (we store `int64`). A lightweight EC2 collector subscribes to the NIFTY and BANKNIFTY near-month futures on the `NSE_FNO` segment, writes one row per server push to daily `zstd` Parquet, and a cron job syncs to S3 after the close. The May contract expired during collection; the loader stitches

the May and June contracts into one continuous front-month series by root symbol, treating the roll (and every overnight gap) as a segment boundary. The result is 21 valid trading days (12 May – 12 June 2026), $\approx 57,000$ events/instrument/day (≈ 1.1 M per instrument, exceeding FI-2010’s ≈ 400 k scale). Figure 1 summarises the end-to-end data-engineering flow.

3.2 Preprocessing and labelling

The pipeline (i) stitches the front-month series, (ii) re-orders Dhan’s columns to the interleaved FI-2010 layout $[P_i^a, V_i^a, P_i^b, V_i^b] \times 10$, (iii) cleans crossed quotes and tick spikes, (iv) segments per trading day so labels and windows never cross a day/contract boundary, (v) z-scores using *training* statistics only, and (vi) forms lazy sliding windows of length $L = 100$. Labels follow the FI-2010 smoothed-mid convention: with m^-, m^+ the mean mid over the k events behind/ahead and $r = (m^+ - m^-)/m^-$, the label is Up if $r > \alpha$, Down if $r < -\alpha$, else Stationary, with a fixed $\alpha = 10^{-5}$ calibrated for the index-futures price scale.

3.3 Microstructure feature engineering

On top of the raw 40-feature vector we engineer microstructure channels: order-flow imbalance (OFI) at level 1 and summed over levels 1–5 (segment-aware), the micro-price $p_{\text{micro}} = \frac{V_1^b P_1^a + V_1^a P_1^b}{V_1^a + V_1^b}$, relative spread, depth imbalance, and the *Dhan-unique per-level order count*. These define feature sets used for ablation: **base40** (raw), **micro** (+6 scalars), **orders60**, **depth80** (20 levels), and **all** (66 features).

3.4 Models

All models share a $(B, 100, 40) \rightarrow \mathbb{R}^3$ contract and are compared under one protocol. We reproduce three baselines (Figure 2): **DeepLOB** (CNN+Inception+LSTM) [8], **MLPLOB** (MLP + Bilinear Normalization, the ablation floor), and **TLOB** (dual-attention transformer, the SOTA reference, vendored faithfully) [1].

The proposed **MambaLOB** (Figure 4) projects the LOB window to d_{model} and applies n selective state-space blocks (Figure 3) realising an input-dependent linear recurrence

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t, \quad (1)$$

where $\bar{A}, \bar{B}, C$ depend on the input (the selection mechanism), giving $O(L)$ time and memory and a parameter count independent of L . **ConvMambaLOB** prepends DeepLOB’s convolutional spatial front-end to the Mamba core, replacing the sequential LSTM with the selective SSM.

3.5 Training and evaluation

Training uses class-weighted cross-entropy, AdamW with cosine decay, and early stopping on validation F_1 . We report weighted and macro F_1 side by side

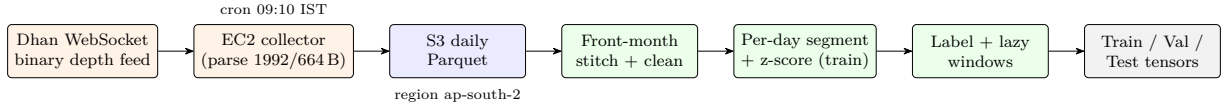


Figure 1: Engineered NSE data pipeline. The reverse-engineered Dhan binary feed (1992 B full / 664 B compact frames) is parsed on an EC2 collector and written as daily Parquet to S3; the modelling layer then stitches the front-month series across the expiry roll, cleans, segments per trading day, z-scores on training statistics, labels, and emits lazily-windowed tensors aligned to FI-2010.

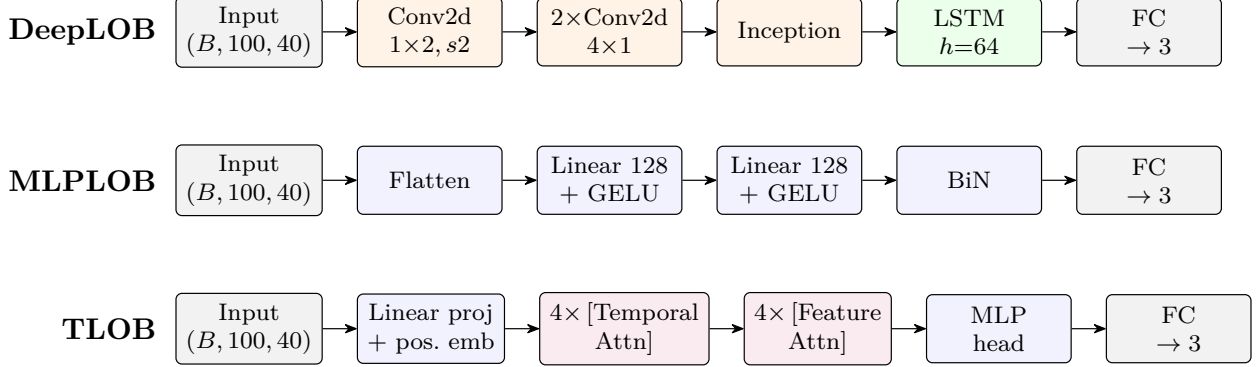


Figure 2: Reproduced baseline architectures, all sharing the $(B, 100, 40) \rightarrow 3$ contract: DeepLOB (CNN+Inception+LSTM), MLPLOB (flatten+MLP with Bilinear Normalization, the ablation floor), and TLOB (alternating temporal and feature self-attention, the transformer SOTA).

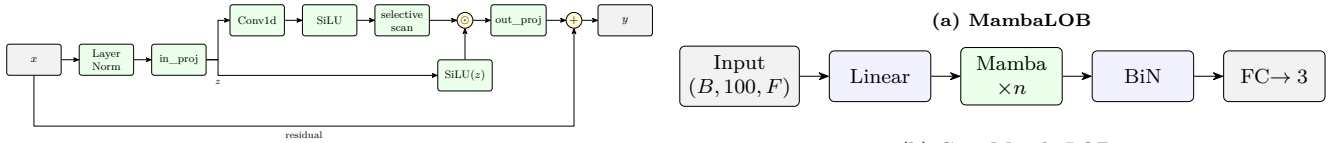


Figure 3: The selective state-space (Mamba) block: the input is normalised and split into two streams; one passes a depthwise causal convolution and SiLU into the selective scan (input-dependent $\bar{A}, \bar{B}, C$), the other forms the SiLU gate. The gated output is projected back and added to the residual; the scan is the only sequential step and runs in $O(L)$.

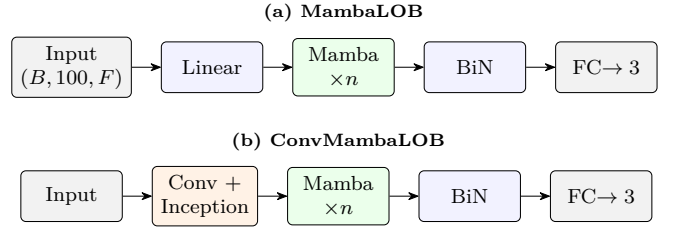


Figure 4: Proposed architectures. MambaLOB applies n selective state-space blocks for $O(L)$ temporal mixing; ConvMambaLOB prepends a convolutional spatial front-end before the Mamba core.

(the FI-2010 headline is a *weighted* F_1 , whereas independent reproductions report *macro*), plus accuracy and Matthews correlation (MCC), against naive majority/stationary/random baselines. The NSE split is chronological (train 12 May–4 June; test 5–12 June).

Table 1: FI-2010 (CF_7, $L=100$): weighted / macro F_1 (%).

Model	$k=10$	$k=100$
DeepLOB	82.5 / 71.9	—
TLOB	88.0 / 81.1	92.5 / 92.5
MambaLOB	79.2 / 67.4	69.3 / 68.7

4 Experiments and Results

4.1 FI-2010 reproduction

Table 1 validates the pipeline: DeepLOB’s weighted F_1 matches the paper’s headline ($\approx 83\%$) and TLOB reaches the SOTA range, confirming faithful reproductions before transfer.

4.2 NSE transfer

On the engineered NSE benchmark (Table 2, Figure 5) all models beat the naive baselines, so learnable signal exists, but absolute scores fall well below FI-2010 — the expected degradation on new data, consistent

with LOBCAST [6]. TLOB leads throughout. The key transfer finding for the proposed model: on NIFTY, MambaLOB (68k parameters) outperforms DeepLOB (191k) and matches MLPLOB, second only to TLOB — whereas on FI-2010 it trailed DeepLOB. The selective SSM thus transfers *better* to the Indian market, at $\approx 1/27$ of TLOB’s parameters.

4.3 Feature engineering and improved architectures

Microstructure feature engineering helps (Figure 6): the all set lifts MambaLOB’s weighted F_1 by up to +0.068

Table 2: NSE Scheme-A test weighted F_1 (range over $k \in \{10, 20, 50, 100\}$). Lift = F_1 minus best naive baseline.

Model	NIFTY	BANKNIFTY	Lift
DeepLOB	0.44–0.52	0.46–0.50	0.03–0.12
MLPLOB	0.46–0.64	0.35–0.47	0.01–0.19
MambaLOB	0.52–0.59	0.45–0.48	0.02–0.17
TLOB	0.60–0.71	0.64–0.72	0.25–0.30

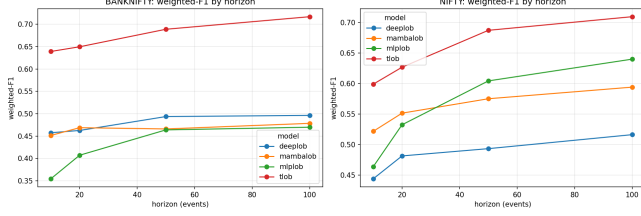


Figure 5: NSE transfer (Scheme A): test weighted- F_1 by horizon. TLOB leads; MambaLOB transfers competitively on NIFTY at a fraction of the parameters.

at $k=100$, with the gain growing with horizon, while deep levels and order counts *alone* do not help in isolation. For architecture (Table 3, Figure 7), ConvMambaLOB is the best of the proposed variants, but no Mamba variant beats TLOB on accuracy. The contribution is the trade-off: **ConvMambaLOB recovers $\sim 90\%$ of TLOB’s quality with $12.5\times$ fewer parameters.**

4.4 Statistical rigour

We re-ran the headline configurations over three seeds and tested each claim with paired across-seed tests and per-sample bootstrap/McNemar tests (Table 4). Microstructure feature engineering is significant by every test ($+0.084$ weighted F_1 , $p=0.001$), and is especially dramatic on BANKNIFTY, where raw features collapse to ≈ 0.45 while engineered features reach ≈ 0.61 . The ConvMambaLOB gain is positive in direction but *not* robust across seeds ($+0.005$, $p=0.25$), and TLOB’s accuracy lead is statistically significant ($p<0.001$). Hyperparameter optimisation gave only a modest gain, and the tuned base MambaLOB — at one layer and $d_{\text{model}}=32$ — matched the tuned ConvMambaLOB, reinforcing that the efficient minimal SSM, not the hybrid, is the sweet spot. The proposed model is already well-calibrated (test ECE 0.041; NLL-optimal temperature $T=1.00$), so its probabilities are usable directly. A class-imbalance study found class-weighted cross-entropy a sound default, with focal loss a marginal alternative at long horizons.

4.5 Efficiency

The motivating advantage of the SSM is its scaling. Figure 8 contrasts TLOB’s $O(L^2)$ attention with MambaLOB’s $O(L)$ scan: MambaLOB’s parameter count is constant in L and its memory/latency grow linearly, while TLOB’s grow quadratically. At $L=800$, MambaLOB is $\approx 31\times$ faster, uses $\approx 3.4\times$ less memory, and has $\approx 750\times$ fewer parameters.

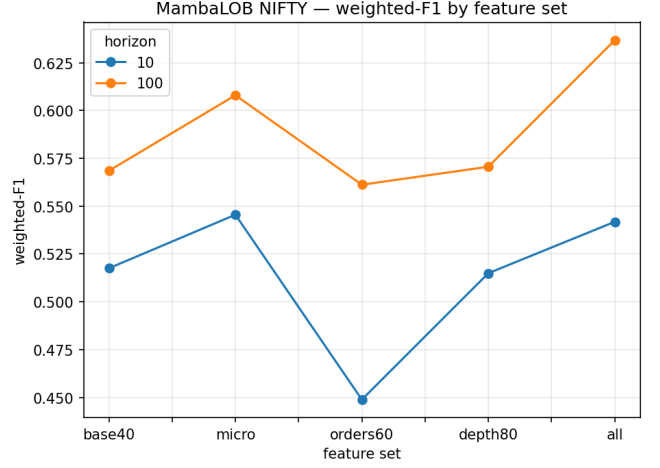


Figure 6: Feature ablation (MambaLOB, NIFTY): weighted F_1 by feature set and horizon. The engineered **micro** and **all** sets sit above the raw **base40**, with the gain widening at the longer horizon; deep levels and order counts in isolation do not help.

Table 3: Architecture comparison (NIFTY+BANKNIFTY, mean over $k \in \{10, 100\}$, **all** features).

Model	Params	wF1	mF1	% TLOB
MambaLOB (base)	70k	0.595	0.508	89%
+ bidirectional	135k	0.597	0.509	90%
+ Conv (ConvMambaLOB)	144k	0.602	0.518	90%
+ both	209k	0.587	0.508	88%
TLOB	1.80M	0.666	0.570	100%

5 Discussion and Conclusion

5.1 Discussion

Interpretation of key findings. Across reproduction, transfer, and a multi-seed rigour study, the consistent story is that the contribution of the selective-SSM models is *efficiency*, not accuracy: a tiny MambaLOB recovers most of the transformer’s quality at $12\text{--}25\times$ fewer parameters and linear-time scaling, but TLOB’s accuracy lead is real and significant. The largest *significant* improvement on the Indian market comes not from architecture but from *microstructure feature engineering* — the engineered channels are what make the weaker NSE signal learnable.

Limitations. The stored feed is a near-regular ≈ 0.4 s snapshot stream rather than tick-by-tick data; the NSE sample spans a single month; and the architecture gains, while directionally positive, are within seed noise. These temper any claim of broad generalisation.

Future work. A spread-relative (transaction-cost) labelling study, a cost-aware backtest with the calibrated probabilities, and walk-forward robustness across more months and instruments.

5.2 Conclusion

We built the first ML-ready NSE index-futures LOB dataset through a full reverse-engineering and cloud

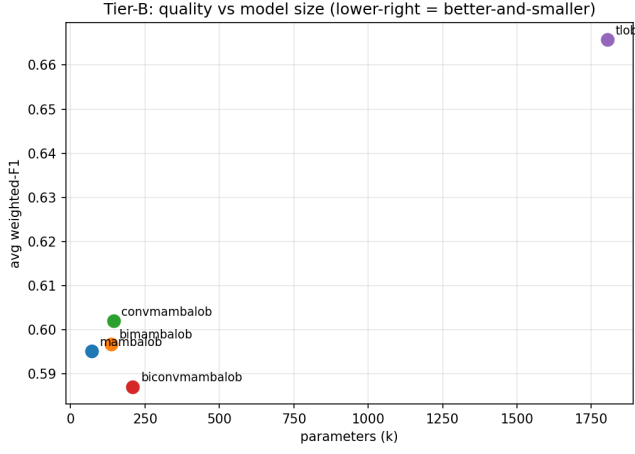


Figure 7: Quality vs. model size. The Mamba family clusters near TLOB’s accuracy at an order of magnitude fewer parameters; the lower-right (better-and-smaller) is the design target.

Table 4: Significance (weighted F_1). Across-seed: paired t -test over 12 cells.

Claim	$\Delta (p)$	Verdict
Features (all vs base40)	+0.084 (0.001)	significant
ConvMambaLOB vs MambaLOB	+0.005 (0.25)	not robust
Mamba/Conv vs TLOB	−0.07 (< 0.001)	TLOB ahead

data-engineering effort, reproduced the standard deep LOB baselines, introduced MambaLOB and ConvMambaLOB, and evaluated everything honestly with multi-seed significance, calibration, and hyperparameter optimisation. The result is a clear efficiency–accuracy trade-off and a significant microstructure-feature effect, providing a reproducible foundation for LOB modelling on Indian index futures.

References

- [1] Leonardo Berti and Gjergji Kasneci. TLOB: A novel transformer model with dual attention for price trend prediction with limit order book data. *arXiv preprint arXiv:2502.15757*, 2025.
- [2] Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.
- [3] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.
- [4] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations (ICLR)*, 2022.
- [5] Adamantios Ntakaris, Martin Magris, Juho Kanninen, Moncef Gabbouj, and Alexandros Iosifidis. Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*, 37(8):852–866, 2018.

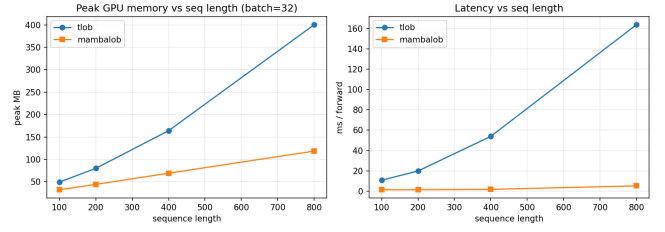


Figure 8: Sequence-length scaling: TLOB ($O(L^2)$) vs. MambaLOB ($O(L)$). The gap widens sharply with L .

- [6] Matteo Prata, Giuseppe Masi, Leonardo Berti, Viviana Arrigoni, Andrea Coletta, Irene Cannistraci, Svitlana Vyetenko, Paola Velardi, and Novella Bartolini. LOB-based deep learning models for stock price trend prediction: A benchmark study. *Artificial Intelligence Review*, 57:116, 2024. arXiv:2308.01915.
- [7] Sasha Stoikov. The micro-price: a high-frequency estimator of future prices. *Quantitative Finance*, 18(12):1959–1966, 2018.
- [8] Zihao Zhang, Stefan Zohren, and Stephen Roberts. DeepLOB: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.